Detection of Terrestrial Bow Shock Crossings in MMS Data and Region Classification Using Neural Network

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***Abstract*—The manual identification of collisionless shock crossings in large-scale space physics datasets is a labor-intensive bottleneck for statistical studies. This work presents the implementation and validation of an automated bow shock detection pipeline developed for NASA's Magnetospheric Multiscale mission, based on recently published methodologies. We first implement the baseline 3D Convolutional Neural Network (CNN) to classify plasma regions from 3D ion energy distributions. Furthermore, we propose and validate a new architecture that combines the 3D VDF data with a parallel branch of 1D plasma features (e.g., plasma beta, entropy, and temperatures). When trained on one month of data and tested on another, our proposed model achieves a classification accuracy of 98.61%, outperforming the 98.35% accuracy of the baseline 3D CNN. This improved classifier is then used to identify bow shock crossings by analyzing the temporal evolution of region classification probabilities. Our work confirms the validity of the original approach and presents a measurable improvement by integrating physical features directly into the deep learning model.**

***Index Terms*—Bow Shock, Collisionless Shocks, Machine Learning, Convolutional Neural Networks (CNN), MMS (Magnetospheric Multiscale Mission), Plasma Physics, Automated Classification, FPI.**

I. Introduction

Collisionless shocks are common phenomena in the study of space and astrophysical plasmas, playing an important role in energy dissipation and particle acceleration as described in [1]. [2] describes The Earth's bow shock, the boundary formed where the supersonic solar wind (SW) is abruptly slowed and heated by the planet's magnetosphere, serves as a natural laboratory for studying these processes .

The [1] mentions Magnetospheric Multiscale (MMS) mission provides plasma and field measurements at unprecedented resolution, making it ideal for investigating these boundaries . However, the sheer volume of data produced by MMS presents a significant challenge. According to [1], [2], identifying bow shock crossings has been a labor-intensive manual process.

This data-rich environment necessitates automated methods. Machine learning (ML), particularly deep learning, has emerged as a powerful tool for classifying massive scientific datasets [2]. This paper details our project to first implement and validate a state-of-the-art ML pipeline for automated bow shock detection based on two key works:

The [2] introduced a 3D Convolutional Neural Network (CNN) to classify 3D ion energy distributions from the MMS Fast Plasma Investigation (FPI) instrument into distinct plasma regions.

The method described in [1], applied this trained classifier to build a large database of MMS bow shock crossings by identifying rapid transitions between the classified regions.

From a computer science perspective, we first focus on the computational implementation of this entire pipeline. We acquire and process public MMS FPI data, design and train the 3D CNN model (hereafter Model-1), and apply the trained classifier to identify bow shock events.

In addition to this implementation, we introduce a new CNN architecture (hereafter Model-2) that integrates 1D plasma features from [3] directly with the 3D VDF data. We demonstrate that this new architecture, which leverages both the raw VDF and calculated physical parameters, achieves a measurably higher classification accuracy when generalizing to unseen data from a different time period.

This paper is organized as follows: Section II provides the theoretical background. Section III summarizes the related work. Section IV details our methodology for data processing and the baseline CNN design. Section V introduces our proposed model, including the exploratory clustering analysis that motivated its design. Section VI describes the bow shock detection algorithm. Section VII presents the comparative performance of both models, and Sections VIII and IX discuss limitations and conclude the work.

II. Theoretical Background

A. Plasma Regions and Ion Distributions

The classification model depends on the fact that different plasma regions traversed by MMS have distinct and identifiable signatures in their 3D ion velocity distribution functions (VDFs) as described in [1], [2]. The four primary dayside regions are: Solar Wind (SW) is The supersonic plasma from the Sun, characterized by a cold, narrow-energy, beam-like ion distribution. Ion Foreshock (IF) is a region upstream of the bow shock, populated by SW ions mixed with ions reflected back from the shock, creating a more complex distribution. Magnetosheath (MSH) is the region between the bow shock and the magnetopause, containing hot, dense, and turbulent plasma. Its VDF is characterized by a broadened, high-energy, and anisotropic distribution. Magnetosphere (MSP) is the region dominated by Earth's magnetic field. Its ion distribution is typically hot, tenuous, and relatively isotropic. The bow shock itself is the boundary separating the upstream (SW, IF) regions from the downstream (MSH) region.

III. Related Work

This project is built directly on the methodologies presented in [2] and [1].

A. Plasma Region Classification

A 3D CNN for automated classification of the FPI/DIS 3D ion distributions was first proposed in [2]. Their model, inspired by the VoxNet architecture proposed in [4], was trained on manually labeled data. A critical contribution was their data processing (log-scaling, normalization, and axis-wrapping) and their use of stratified sampling to create a balanced training set from the highly imbalanced source data. Their final model achieved over 98% accuracy.

B. Bow Shock Database Compilation

This work was extended by [1] to the specific task of bow shock detection. They used the trained CNN as a region classifier and applied it to over 5 years of MMS data [5]. Their key innovation was an algorithm to identify shock crossings based on the classifier's output probabilities. They defined a probability difference parameter, Δp, to separate upstream (SW/IF) and downstream (MSH) regions. By finding sharp extrema in the temporal derivative of this Δp signal, they could automatically flag shock transitions.

IV. Methodology

The methodology of this work involved creating the complete identification pipeline for bow shock crossing from [1] and attempting to construct a new model showing improvements over model described in [2].

A. Data Acquisition and Preprocessing

Level-2 FPI/DIS data (dis-dist) was acquired from the MMS Science Data Center public portal [5]. This data, stored in Common Data Format (CDF) files, contains the [32, 16, 32] ion distribution arrays.

we followed a 4-step preprocessing procedure as described in [2]was applied to each sample: Zero Replacement: All zero values (representing no counts) were replaced with the minimum non-zero value in the array. Logarithmic Scaling: The log_{10} of each element was computed to handle the large dynamic range of particle fluxes. Normalization: The array was normalized to a [0, 1] range by subtracting the minimum and dividing by the new maximum. Axis Wrapping: The array was wrapped along the third axis (azimuthal angle) by 16 elements. This step is crucial as it centers the solar wind beam, which appears at the edges of the detector's field of view.

B. MMS and Data Products

The models use data from two primary MMS instruments. The Fast Plasma Investigation (FPI) suite provides the two main inputs. First, its Dual Ion Spectrometer (DIS) measures the 3D ion VDF, producing the [32, 16, 32] (energy, polar, azimuthal) array that serves as the volumetric input to the 3D CNN [2].

Second, the FPI's ion (DIS) and electron (DES) spectrometers provide 1D moment data (ion/electron density and temperature). We combine these moments with magnetic field data from the FluxGate Magnetometer (FGM) to form an array of 8 features (β, s, n_i, n_e, $T_{\perp i}$, $T_{\parallel i}$, $T_{\perp e}$, $T_{\parallel e}$) used in our extended model.

C. Dataset Curation and Sampling

The full dataset is both massive and highly imbalanced. As mentioned in [2], training on the full dataset is impractical and can overfit the model towards the regions having more instances making the model biased. To address this we have used the methodology described in [2] to sample the data. A training subset was created by randomly selecting samples from each of the four classes. To create a balanced set, smaller selection coefficients were used for abundant classes (SW, MSH) and larger coefficients for rare classes (IF, MSP). This ensures the model learns to identify all regions effectively.

D. 3D CNN Model Architecture (Model-1)

A 3D CNN architecture in [2] (which was inspired by VoxNet [4]), was implemented using the Keras and TensorFlow frameworks. This is our baseline Model-1.

E. Model Training

The model was trained on a dataset of 24266 samples which was compiled as a balanced subset of the human labeled dataset given in [2]. We have used the Adam optimizer and the categorical cross-entropy loss function, as specified in [2].

V. Model Enhancement with 1D Features

While the 3D CNN (Model-1) is effective, it only uses the 3D VDF. We hypothesized that classification accuracy could be improved by also providing the model with a set of 1D plasma features mentioned in [3]

A. New Features

Statistical studies, such as [3], have shown that plasma regions can be clearly distinguished by their 1D moment properties. We used 8 such physically-significant features from the FPI and FGM data: Ion/Electron Density (n_i, n_e) Ion/Electron Temperatures
($T_{para\_i}, T_{perp\_i}, T_{para\_e}, T_{perp\_e}$) Plasma Beta ($\beta$): The ratio of thermal pressure to magnetic pressure. Ion Specific Entropy (s): ($s = T_i/n_i^{2/3}$) that is very low in the Magnetosheath and very high in the Magnetosphere as mentioned in [3].

B. Exploratory Analysis: Feature Clustering

To validate this hypothesis, we first performed an exploratory analysis. We created a combined dataset with 16,392 features per sample (the 16,384 flattened VDF features + 8 features). An unsupervised clustering algorithm, K-Means was applied to this high-dimensional space and was run on the full training set. The algorithm achieved a cluster purity score of 0.5410 for K-Means with cluster distribution $\{0 : 17598, 1 : 2243, 2 : 1129, 3 : 2315\}$. The cluster labels are arbitrary denoting a cluster. The analysis indicated that we should resort to methods such as sophisticated feature embedding or supervised learning for classification.

C. Model-2: Proposed CNN-Feature Network

To leverage these 8 features, we designed the architecture shown in Fig. 1. This branched model processes both data types in parallel. The CNN branch processes the 3D VDF Input(32x16x32x1) through the same convolutional and pooling layers as Model-1, resulting in a 1D feature vector from a Dense(128) layer. The Feature branch processes the 8-feature 1D vector through two dense layers, Dense(32, ReLU) and Dense(16, ReLU). The 128-unit vector from the CNN branch and the 16-unit vector from the Feature branch are then concatenated into a single 144-unit vector. This combined vector is passed through a final Dense(128, ReLU) layer and a Dense(4, Softmax) output layer.

VI. Bow Shock Detection Algorithm

With a trained classifier, the project's focus shifted to recreating the bow shock detection algorithm from [1].

A. Region Classification

The trained CNN [2] used as an inference engine, applied to the full time-series of preprocessed FPI/DIS data. For each 4.5s measurement interval, the model outputs a vector of four probabilities: p_{SW}, p_{IF}, p_{MSH}, and p_{MSP}, which sum to 1.

B. Probability Difference Calculation

The bow shock is the boundary between the upstream regions (SW and IF) and the downstream region (MSH). To quantify this transition, [1] defines a probability difference parameter, Δp:

$$\Delta p = (p_{SW} + p_{IF}) - p_{MSH} \tag{1}$$

When the spacecraft is in the solar wind or foreshock, $p_{SW} + p_{IF} \approx 1$ and $p_{MSH} \approx 0$, yielding $\Delta p \approx +1$. Conversely, in the magnetosheath, $p_{MSH} \approx 1$, yielding $\Delta p \approx -1$.

C. Event Identification

A time-series of Δp values was generated. The detection algorithm described in [1] was then implemented. To avoid noisy fluctuations and ambiguous mixed regions [2], all data points where $|\Delta p| < 0.9$ were filtered out [1]. A moving median filter was applied to the Δp time-series to smooth out high-frequency noise. A shock crossing is a rapid transition from $\approx +1$ to ≈ -1 (inbound) or ≈ -1 to $\approx +1$ (outbound). This transition is identified by calculating the discrete temporal derivative, $d(\Delta p)$:

$$d(\Delta p)_i = \Delta p_{i+1} - \Delta p_i \tag{2}$$

Large local minima in $d(\Delta p)$ (sharp negative spikes) correspond to inbound crossings, while large local maxima (sharp positive spikes) correspond to outbound crossings. These extrema are flagged as candidate bow shock events.

A conceptual plot of this detection method is shown in Fig. 3.

VII. Results and Comparison

A. Classification Accuracy of Model-1

The performance of the recreated 3D CNN model (Model-1) was evaluated on a hold-out test set. The model achieved a classification accuracy comparable to the benchmark of $> 98\%$ reported in [2]. This result validates the model architecture and the data preprocessing and sampling strategy.

As noted in [2], the most significant source of error was the misclassification of Ion Foreshock (IF) samples, which were most often confused with Solar Wind (SW). This is an expected and physically reasonable confusion, as the IF is, in definition, a mixture containing SW ions.

B. Bow Shock Candidate Database (from Model-1)

The detection algorithm described in Section VI was applied to several months of MMS data using the trained Model-1. This process generated a large list of candidate bow shock crossings, each tagged with a precise epoch.

For each candidate event, associated data was programmatically compiled.

The epoch of the crossing. The spacecraft position in Geocentric Solar Ecliptic (GSE) coordinates. The availability of high-resolution burst mode data for the event.

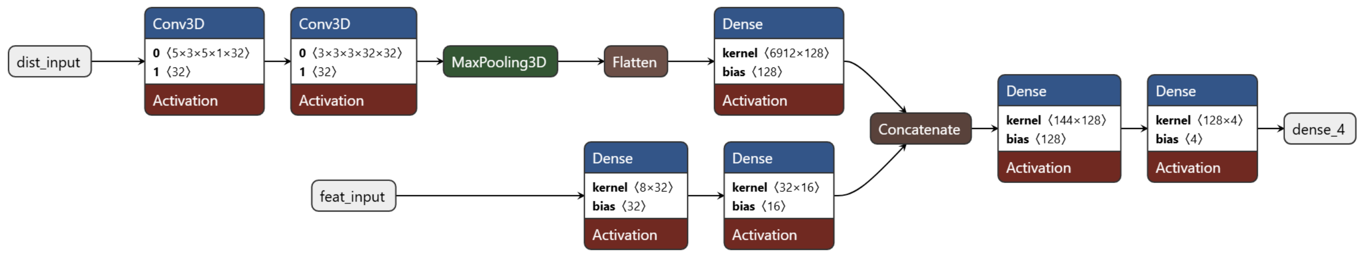

Fig. 1. Architecture of Model-2, our proposed network. It combines a 3D CNN branch for VDFs with a 1D Dense branch for the 8 plasma features.

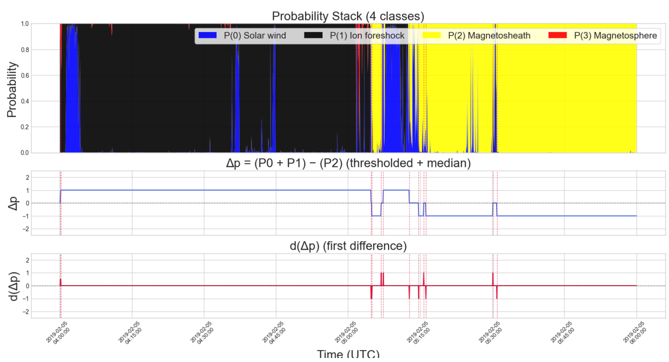

Fig. 2. Probability stack and shock detection analysis. The top panel shows the CNN class probabilities for different plasma regions(SW-Blue, IF-Black, MSH-Yellow, MSP-Red), while the middle and bottom panels show Δp and its first derivative $d(\Delta p)$, respectively.

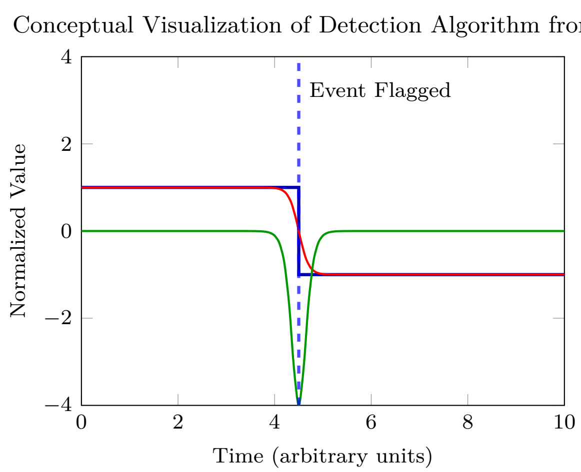

Fig. 3. Representation of the detection algorithm described [1]. The CNN probability output (blue) differentiates upstream and downstream regions, the probability difference Δp (red) shows the transition, and its derivative $d(\Delta p)$ (green) highlights the detected shock boundary.

Upstream solar wind parameters (e.g., B, V_{SW}, M_A) from the OMNI database [6], time-shifted to the MMS location.

This structured database forms the primary output of this recreation project. The spatial distribution of the identified candidate events, when plotted in the GSE X-Y plane, correctly traces the expected statistical location of

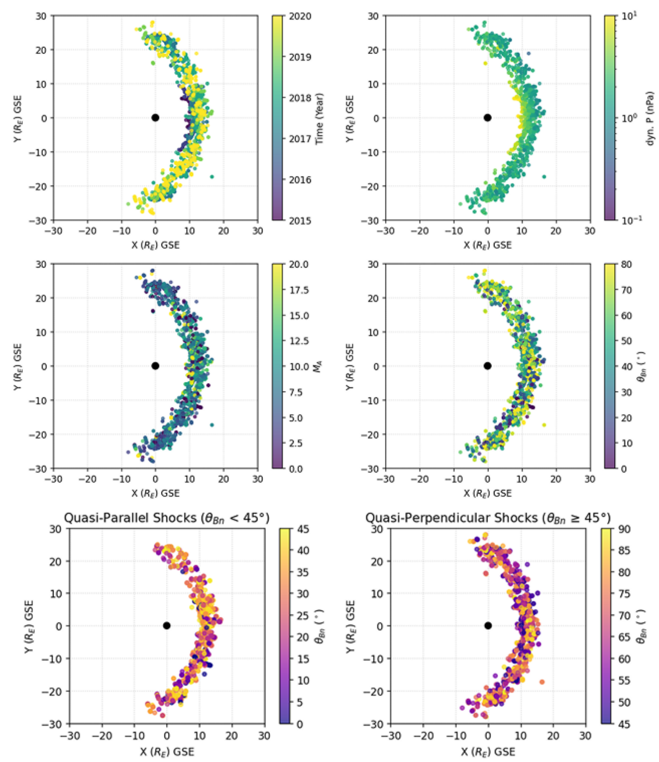

Fig. 4. Location of the crossings of all of the shocks in the database projected on the ecliptic plane and normalized to Earth radius. Color bars in each panel represent time(top-left), dynamic pressure(top-right), MA(middle-left), and θBn(middle-right) along with Quasi-Parallel(bottom-left) and Quasi-Perpendicular(bottom-right) Shock Positions

the Earth's bow shock, consistent with the findings in [1] (Fig. 7).

C. Comparative Model Performance

To test the generalization capabilities of our proposed model (Model-2) against the baseline (Model-1), we followed a validation strategy: both models were trained on the balanced dataset from December 2017 and evaluated on the balanced, unseen dataset from November 2017.

Both models performed well, but Model-2 showed a clear improvement in overall accuracy and in specific class performance.

TABLE I
PER-CLASS ACCURACY COMPARISON (TEST SET: NOV 2017)

Class	Region	Model-1 Acc.	Model-2 Acc.
0	Solar Wind	99.21%	98.74%
1	Ion Foreshock	96.39%	96.61%
2	Magnetosheath	99.40%	99.40%
3	Magnetosphere	98.22%	99.76%
	Overall Accuracy	**98.35%**	**98.61%**

TABLE II
CONFUSION MATRICES (ROWS=TRUE, COLS=PREDICTED)

Model	True Class	Predicted Class 0 (SW)	1 (IF)	2 (MSH)	3 (MSP)
Model-1	0 (SW)	7831	62	0	0
	1 (IF)	145	5481	43	17
	2 (MSH)	0	29	4819	0
	3 (MSP)	0	95	9	5735
Model-2	0 (SW)	6825	87	0	0
	1 (IF)	113	5493	80	0
	2 (MSH)	0	29	4819	0
	3 (MSP)	0	9	5	5825

Model-1 (Baseline 3D CNN): Achieved an overall accuracy of 98.35%. Model-2 (Proposed CNN): Achieved a superior accuracy The per-class accuracy, detailed in Table I, shows that Model-2's primary advantage comes from a significant improvement in identifying the Magnetosphere (Class 3), boosting its recall from 98.22% to a near-perfect 99.76%. The confusion matrices in Table II confirm this; Model-2 reduced the number of Class 3 samples misclassified as Class 1 (Foreshock) from 95 to just 9.

This confirms our hypothesis: the new features provide macroscopic context that helps the model distinguish between regions.

VIII. Discussion and Limitations

This project successfully recreated the computational pipeline for automated bow shock detection. The results validate that the 3D CNN approach described in [2] is a robust foundation for the probability-based detection algorithm [1]. Furthermore, our development of Model-2 demonstrates that this already-high accuracy can be improved. The new approach shows that while 3D VDFs contain most of the information, they are not all of it. The 1D plasma properties provide macroscopic context that the 3D CNN cannot easily learn, particularly for distinguishing the Magnetosphere. However, limitations remain: Misidentifications: The algorithm is not infallible. As noted in [1], the classifier can be fooled by other plasma boundaries that mimic a shock transition, such as hot flow anomalies (HFAs).

Manual Verification is Essential: This project generated a candidate database. The original authors of [1] visually check each shock to filter misidentifications. This recreation confirms that the automated output is a first pass and that a human-in-the-loop verification step remains indispensable.

IX. Conclusion and Future Work

This paper presented a successful implementation and validation of the CNN-based methodology for automated bow shock detection from MMS data. We first built a baseline 3D CNN replicating the work of [2]. We then proposed and built a neural network (Model-2) that integrates 1D plasma features with the 3D VDFs.

When tested on an unseen month of data, our Model-2 (98.61%) outperformed the baseline Model-1 (98.35%), demonstrating a more robust and accurate classification. This project validates the original methodology as a powerful tool for accelerating research in space physics and offers a clear path for future model enhancement.

Future work based on these findings includes:

Performing the large-scale manual, visual verification of all candidate events generated by our superior Model-2 to create a finalized, clean database. Further enhancing the model by exploring attention mechanisms to allow the model to weigh the importance of the CNN branch vs. the feature branch. Implementing a temporal model (e.g., a ConvLSTM) that takes a sequence of 3D VDFs (e.g., data from t and $t-1$) as input to better classify ambiguous boundary regions. Applying the detection algorithm from our trained Model-2 to the full 5+ year MMS dataset to generate a new, comprehensive bow shock database.

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